

MUG Hackathon: MongoDB x Tavily x LastMile AI

Project: System Maintenance Agent

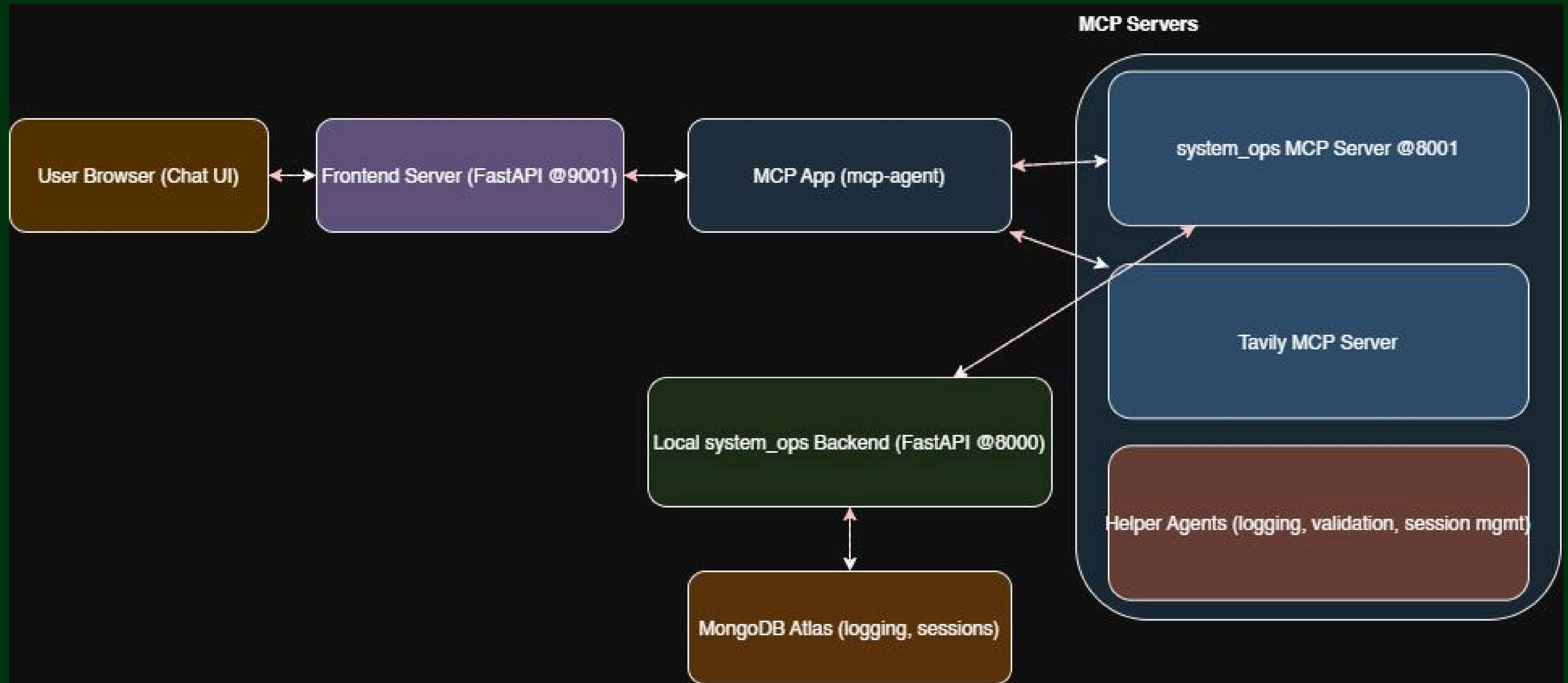
MCP Powered agent for system maintenance/
diagnostic tests etc.

Project overview

This project, developed for the MongoDB User Group Hackathon, aims to create an agent powered by the Model-Context-Protocol, to mimic a remote technician and perform diagnostics, application installs and solve other system related issues.

The application uses the MCP protocol for implementing the agent, Tavily MCP for implementing search functionality for user queries and MongoDB atlas for logging and session related data storage.

Project Architecture



Tech stack:

- Language & runtime: Python 3.13
- Web framework: FastAPI + Uvicorn (system_ops backend and frontend server)
- Agent framework: mcp-agent (MCPApp, AgentSpec, workflows)
- LLM: Google Gemini via GoogleAugmentedLLM
- MCP protocol: system_ops via streamable_http, Tavily via remote MCP (mcp-remote)
- Windows ops layer: PowerShell + WMI
(Get-CimInstance, Get-Process, Get-Service, Start-MpScan, winget/Chocolatey)
- Data & config: MongoDB Atlas (pymongo) for session logs, python-dotenv
+ .env for local configuration

Use Cases

The project can have the following use cases:

- a. Helping enterprises automate systems maintenance for workforces
- b. Helping companies provide better technical support via agents.
- c. Helping people perform system diagnostics relatively easily.
- d. Auditing and compliance for support actions through logging.

Project Status

Complete	Basic Prototype for executing commands within windows environment
Complete	MCP plus Helper Agent setup for powershell command execution, app installs, system diagnostics etc.
Complete	Database Logging for per session actions
Complete	Interactive front-end with Human-in-the-loop execution
Future Scope	Support for more MCP servers and agents
Future Scope	Support for Mac and Linux based platforms.
Future Scope	Remote deployment of MCP agent to cloud and remote execution to powershell using webRTC
Future Scope	Separate profile and power-tools for advanced users.

Thank You
